



Getting Started with HTML5

Learning Objectives

- Understand the new features and capabilities of HTML5 and CSS3.
- Learn the fundamental layout of an HTML page, including the head and body sections.
- Write a basic "Hello World" webpage to grasp HTML syntax and structure.
- Use comments to enhance code readability.
- Differentiate between elements and tags.
- Group HTML elements using the div tag for better organization.

Prerequisites

- Basic understanding of programming concept
- Basic understanding of Internet
- Understanding how browser works



What is HTML and How the Web Page Works

Introduction to HTML5

What is HTML?

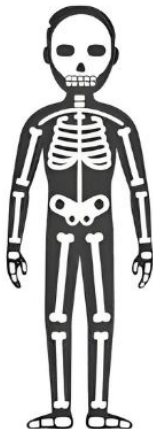
- HTML stands for Hyper Text Markup Language
- HTML is the standard markup language for creating Web pages
- HTML describes the structure of a Web page
- HTML consists of a series of elements
- HTML elements tell the browser how to display the content
- HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.
- HTML5 is the latest version, introducing new features and improvements.

HTML5, CSS and JavaScript ?

HTML



HTML the Skeleton



CSS



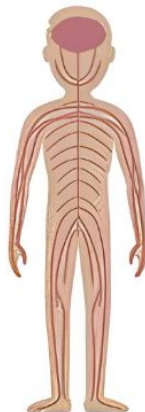
CSS the Skin



JS



Javascript the Brain



Features of HTML5



VS Code setup for HTML5

HTML



Setup Live Server



Live Server

[Ritwick Dey](#) |  74,733,845 installs | ★★★★★ (513) | Free

Launch a development local Server with live reload feature for static & dynamic pages

[Install](#)

[Trouble Installing?](#) 

Hands-On Demo | What is HTML and How the Web Page Works

Pop Quiz

Q. What does HTML stand for?

A

HyperText Markup Language

B

HyperText Markdown Language

Pop Quiz

Q. What does HTML stand for?

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HTML Page Structure (Build First)

HTML with Emmet

- Emmet is a toolkit that lets you write HTML faster using short abbreviations that expand into full code automatically.

`div>ul>li*3`

Emmet Abbreviations

```
<div>
  <ul>
    <li></li>
    <li></li>
    <li></li>
  </ul>
</div>
```

Expanded HTML



Emmet

Basic structure of HTML

```
<!DOCTYPE html> ← Tells the document type
<html> ← The Root Element
  <head> ← Contains the header information
    <title>Title of the Page</title> ← Defines Title of the Page
  </head>
  <body> ← Holds the Content of the Page
    Tags related to layout and formatting
  </body>
</html>
```

Fundamental Structure of an HTML Page

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title> Home Page </title>
</head>
<body>
  <div>
    <h1> Hello World</h1>
  </div>
</body>
</html>
```

Comments in HTML

In HTML, comments are used to add notes or annotations within the code that are ignored by the browser when rendering the webpage.

- **Single-line Comments:** Use `<!-- -->` to create single-line comments. Anything between `<!--` and `-->` will not be displayed in the browser.
- **Multi-line Comments:** HTML doesn't have specific syntax for multi-line comments like some programming languages, but you can achieve multi-line comments by using multiple single-line comments.

A dark-themed code editor window with three colored window control buttons (red, yellow, green) at the top left. It contains two examples of HTML comments and an input tag.

```
<!-- Ask for one email address -->
<input type="email" />

<!-- Ask for multiple email addresses
      (comma-separated) -->
<input type="email" multiple />
```

Q. What tag is used to define the root of an HTML page?

A

`<!DOCTYPE html>`

B

`<html> </html>`

Q. What tag is used to define the root of an HTML page?

A

`<!DOCTYPE html>`

B

`<html> </html>`

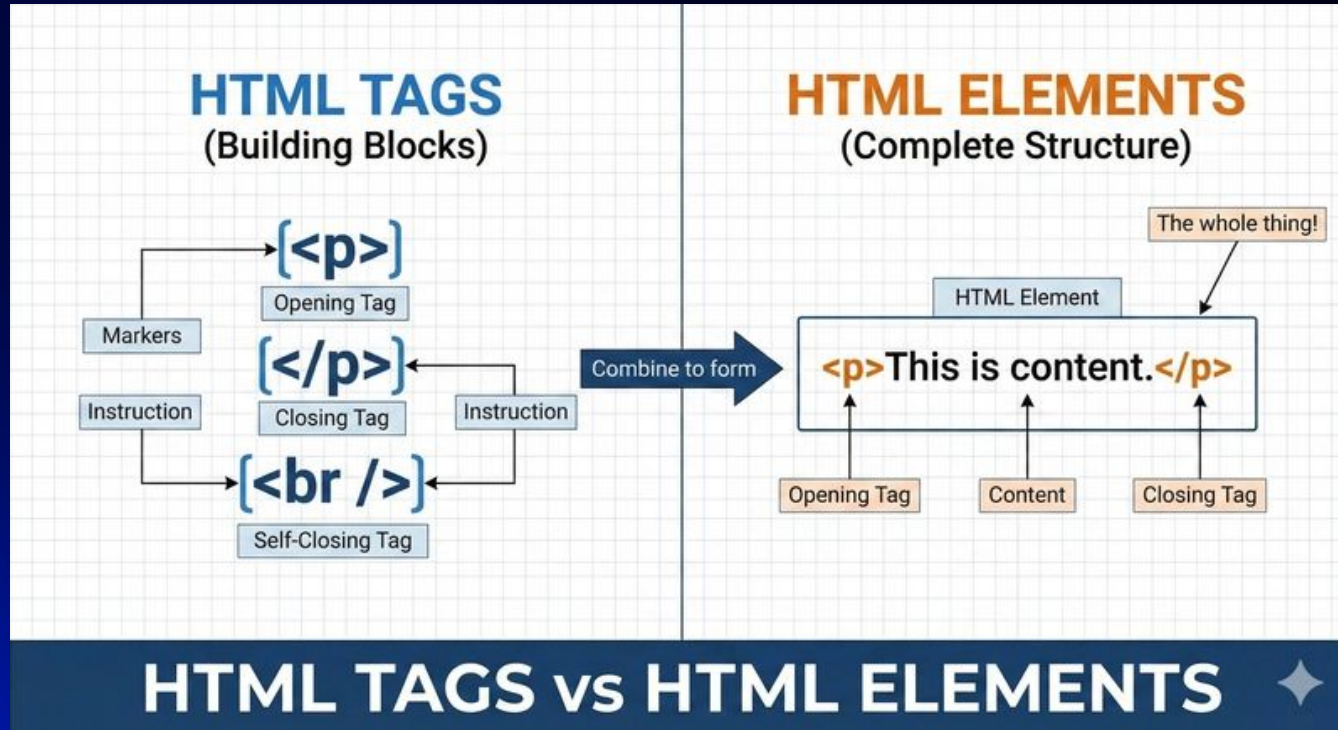
Hands-On Demo | HTML Page Structure (Build First)

Building the Class Project (Home Page)

Fundamental Structure of an HTML Page

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Portfolio</title>
</head>
<body>
  <div>
    <h1>My Portfolio</h1>
    <br>
    <p>My name is John Doe and I am a web developer.</p>
    <hr>
  </div>
</body>
</html>
```

HTML Tags vs HTML ELEMENTS



Organizing Content with `<div>` Tag

The `<div>` tag in HTML is a versatile and essential element used to group together and organize content within a web page. It stands for "division" or "divider" and does not have any specific meaning or styling associated with it by default.

Purpose of `<div>` Tag:

- **Structural Organization:** The primary use of `<div>` is to divide or structure different sections of a web page logically.
- **Styling and Layout:** `<div>` tags provide hooks for applying CSS styles and classes. They allow developers to apply consistent styling across similar sections of a web page

Organizing Content with <div> Tag

```
<div class="container">
  <div class="header">Header Content</div>
  <div class="main-content">
    <div class="sidebar">Sidebar Content</div>
    <div class="article">Main Article Content</div>
  </div>
  <div class="footer">Footer Content</div>
</div>
```

Hands-On Demo | Building the Class Project (Home Page)

Q. Which HTML tag is used to define metadata about an HTML document, such as character encoding and viewport settings?

A

<body>

B

<meta>

Pop Quiz

Q. Which HTML tag is used to define metadata about an HTML document, such as character encoding and viewport settings?

A

`<body>`

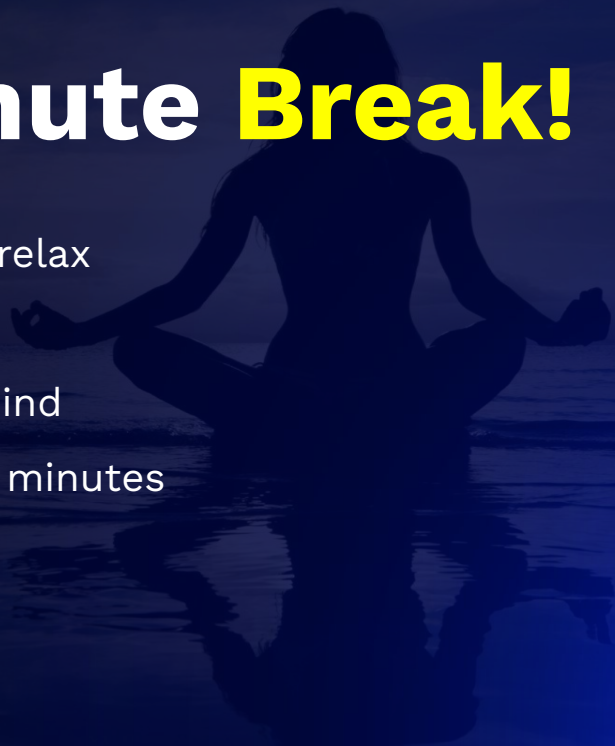
B

`<meta>`



Take A 5-Minute **Break!**

- Stretch and relax
- Hydrate
- Clear your mind
- Be back in 5 minutes



Text Formatting and Lists

Common Formatting Tags

In HTML, formatting tags are used to style text and add meaning to content beyond plain text. Here are some common tags and their uses:

Tag	Purpose
<code></code> and <code></code>	To Bold Text
<code><i></code> and <code></code>	To italic Text
<code><sub></code>	Defines subscripted text
<code><sup></code>	Defines superscripted text
<code><small></code>	Defines smaller text
<code><big></code>	Defines bigger text
<code><ins></code>	Defines inserted text
<code></code>	Defines deleted text
<code><mark></code>	Defines marked/highlighted text

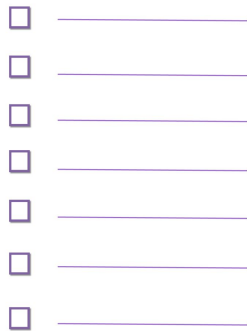
Lists in HTML

HTML lists allow web developers to group related items together. They're like the organizational superheroes of web content. There are three main types of lists in HTML:

Ordered Lists (): Lists where each item is numbered or ordered sequentially.

Unordered Lists (): Lists where each item is marked with a bullet point or similar marker.

Definition Lists (<dl>): Lists that consist of terms and their corresponding definitions.

- 
- A visual example of an unordered list, showing seven items, each consisting of a small square bullet point followed by a horizontal line representing text.
- _____
 - _____
 - _____
 - _____
 - _____
 - _____
 - _____

Ordered List

These lists are all about order and discipline. Each item gets a number (or sometimes a letter) to keep things in line.

How they work:

- Start with the `<ol>` tag to define your ordered list.
- Each list item begins with the `<li>` tag.
- The list items are automatically numbered (1, 2, 3, and so on).

```
<ol>
  <li>Name</li>
  <li>Address</li>
  <li>Phone Number</li>
</ol>
```



```
1. Name
2. Address
3. Phone Number
```

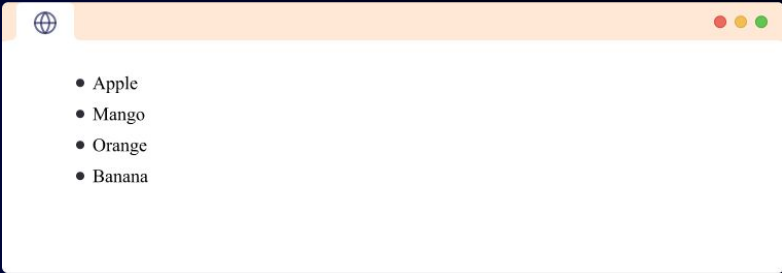
Unordered List

These lists are more laid-back. They use bullet points (small black circles) to mark each item.

How they work:

- Start with the `<ul>` tag for your unordered list.
- Again, each list item is wrapped in an `<li>` tag.

```
<ul>
  <li>Apple</li>
  <li>Mango</li>
  <li>Orange</li>
  <li>Banana</li>
</ul>
```

- 
- A screenshot of a web browser window with a light orange header bar. The browser's address bar shows a globe icon. The main content area displays an unordered list with four items: Apple, Mango, Orange, and Banana, each preceded by a small black circle.
- Apple
 - Mango
 - Orange
 - Banana

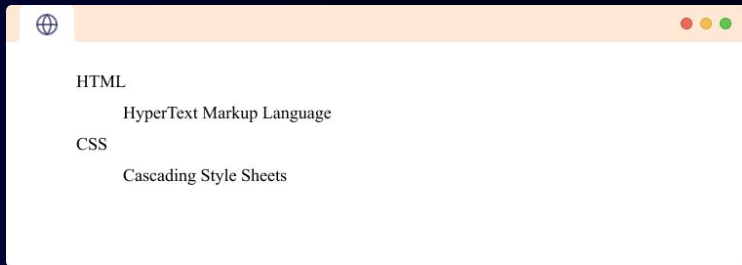
Definition Lists

These are a bit different. They consist of terms and their corresponding definitions.

How they work:

- Use the `<dl>` tag to define your description list.
- `<dt>` defines the term (name), and `<dd>` describes each term.

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>
```



HTML Attributes and Inline Styles



HTML attributes provide additional information about HTML elements and are defined within the opening tag of an element. They can control various aspects of an element's behavior and appearance.

Class (class): Specifies one or more class names for an element, which can be used to apply CSS styles.

ID (id): Specifies a unique identifier for an element, typically used for scripting or styling purposes.

Href (href): Specifies the URL of the linked resource in `<a>` elements.

Style (style): Allows inline CSS styles to be applied directly to an element.

Inline vs Block-level Elements

Block Elements	Inline Elements
Block elements always start from a new line.	Inline elements never start from a new line.
Block elements cover space from left to right as far as it can go.	Inline elements only cover the space as bounded by the tags in the HTML element.
Block elements have top and bottom margins.	Inline elements don't have a top and bottom margin.
Examples of block elements - <code><p></code> , <code><div></code> , <code><hr></code> .	Examples of inline elements - <code></code> , <code>
</code>

Inline vs Block-level Elements



INLINE ELEMENT

```
<p>This is a <span>highlighted</span> word in a paragraph.</p>
```

BLOCK ELEMENT

```
<div>  
<h1>Main Heading</h1>  
<p>This is a paragraph inside a div.</p>  
</div>
```

Hands-On Demo | Text Formatting and Lists

Q. Which tag is used to Strikethrough text?

A

`<s>`

B

`<strike>`

Q. Which tag is used to Strikethrough text?

A

`<s>`

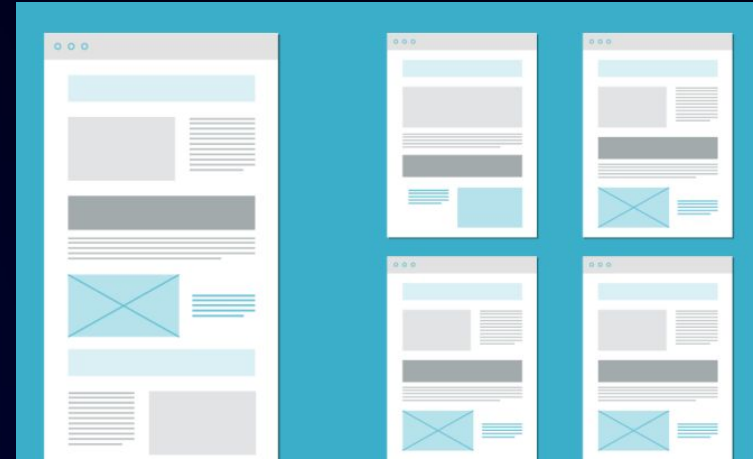
B

`<strike>`

Links and Multiple Pages

Importance of Multiple Pages

- **Better Organization** – Content is divided into sections, making the website easier to manage and understand.
- **Improved User Experience** – Users can find information quickly without scrolling through one long page.
- **Faster Performance** – Smaller pages load faster than a single large page.

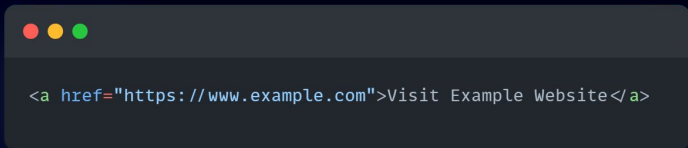


Anchor Tags

Anchor tags (also known as `<a>` tags) in HTML are used to create hyperlinks. They allow you to link to other web pages, documents, or specific sections within the same page. Here's a brief explanation:

`<a>` (Anchor) Tag:

- The `<a>` tag defines a hyperlink.
- It requires an `href` attribute (the URL or destination) to specify where the link leads.
- Example:
`<a href="https://www.example.com">Visit Example</a>`

A dark-themed code editor window with a title bar containing three colored dots (red, yellow, green). The code inside is `<a href="https://www.example.com">Visit Example Website</a>`.

```
<a href="https://www.example.com">Visit Example Website</a>
```

Q. Which attribute specifies the destination of a link?

A

path

B

href

Pop Quiz

Q. Which attribute specifies the destination of a link?

A

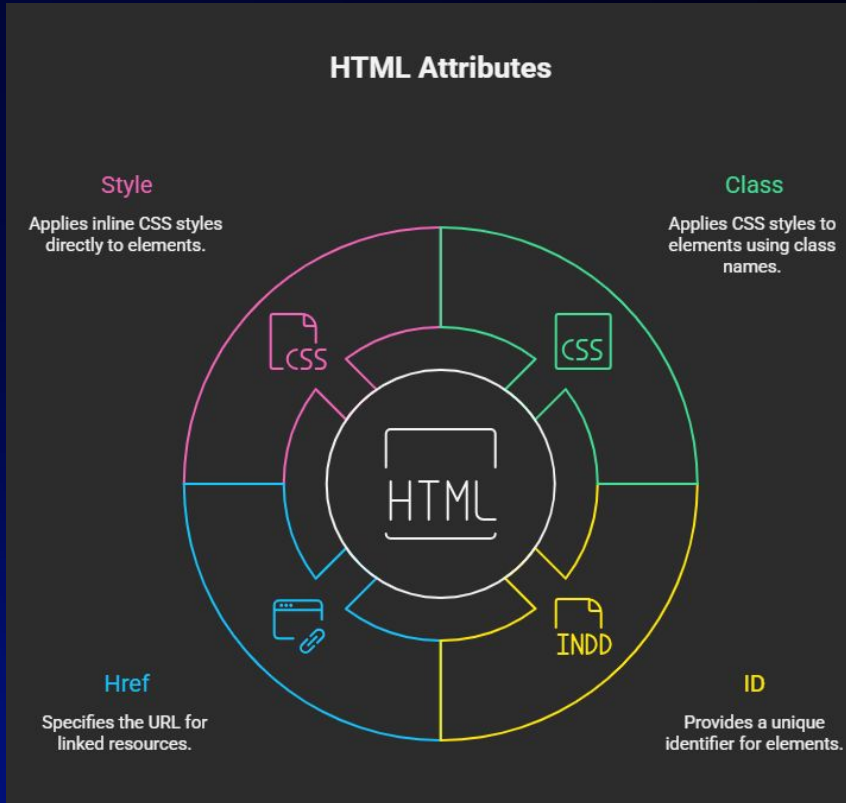
path

B

href

Inline Styling overview

HTML Attributes



Inline Styling

```
<html>
<head>
<body>
<h1>How to add CSS</h1>
<p style="font-size:20px; color:red">This is paragraph</p>
</body>
</html>
```



Inline CSS for `<p>` tag

Here we apply CSS on that line where html element (`<p>`) are use so this type of css is called Inline css

Q. Which attribute is used for inline styling?

A

style

B

CSS

Pop Quiz

Q. Which attribute is used for inline styling?

A

style

B

CSS

Hands-On Demo | Inline Styling

Summary

In this lecture, we have covered:

- What HTML is and how web pages work
- Overview of HTML page structure
- Building a home/landing page HTML project
- Text formatting and lists
- Linking multiple pages
- Inline styling

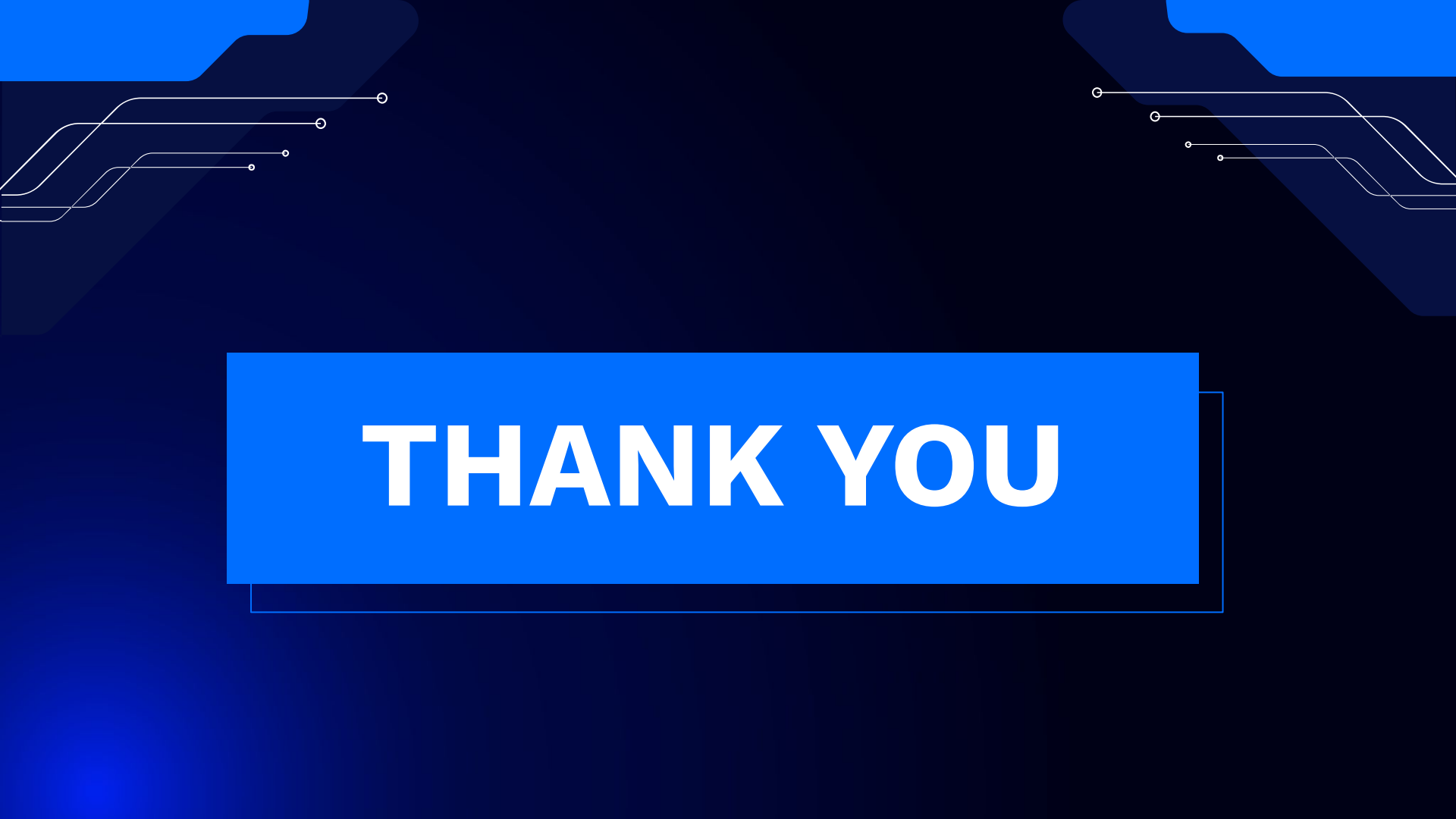




DOUBT RESOLUTION

Important

- Complete the post-class assessment
- Complete assignments (if any)
- Practice the concepts and techniques taught in this session
- Review your lecture notes
- Note down questions and queries regarding this session and consult the teaching assistants.



THANK YOU